

Danielle Schwartz

Systems Engineer Passionate About Bridging Engineers & Scientists to Improve Marine Technology

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EDUCATION

Boston University, College of Engineering, Master of Science in Systems Engineering

Sept. 2024 – May 2027

Concentration: Controls | **GPA: 3.467**

- Relevant coursework: Signals & Systems, Control Systems, Stochastic Simulation, Computer Algorithms (Data Structures & Applications), Optimization Theory, Circuit Theory, Deep Learning, Probability & Statistics, Applied Software Engineering, Machine Learning, Oceanography, Climate History, Advanced Ocean Sciences

University of Arizona, Bachelor of Science in Geosciences

May 2023

Concentration: Earth, Oceans, and Climate | **GPA: 3.94** | Honors: Summa Cum Laude, Graduation with Honors

SKILLS

Languages: Python, PyTorch, MATLAB, C++, C, Rust, Java, HTML/CSS, Bash

CAD: SolidWorks, Onshape, Creo

Robotics & Embedded Systems: ROS2, OpenCV, RViz, UART, Arduino, I2C

Hardware: KiCAD, Soldering, Lathe, Drill Press, Laser Cutter

Tools & DevOps: GitHub, Git, Docker, Linux

Cloud Computing: AWS (SQS, Lightsail)

Fieldwork: Water Sampling, Site Visits, Troubleshooting, System Performance Evaluation, Safety Standards

PROJECTS

Agrobot V2

January 2026 – Present

MassRobotics Form & Function Robotics challenge - First Place Winner + Audience Choice

- Leadership position: **Membership Engagement Lead, Outreach Coordinator, & Safety Officer**
- Coordinated with (Maxon, Mitsubishi, Maeden) sponsors for hardware and software donations according to budget
- Worked in team of 8 engineers to build computer vision algorithm for a 7-DOF tomato picking robot
- Prioritized safety of team members and conference by creating daily safety operating procedures

Plastic Polymer Classification with Deep Learning Models

Spring 2026

- Developed a Deep Learning Framework identify 7 chemically distinct polymer classes (RIC #1–7), utilizing Vision Transformers (ViT) and ResNet-50
- Optimized Feature Extraction by manipulating embedding space vectors to increase Euclidean distance between similar data classes, achieving finer classification precision for high-density localized datasets
- Conducted comparative performance Analysis between transformer and CNN-based spatial filtering, evaluating models via Cross-Entropy Loss and confusion matrices to determine the most robust architecture for noisy sensor data

ANGLER ROV

Summer 2025

- Created low cost, open source ROV with additional, creative designs
- Iterated in team of two through design challenges and physical limitations to deploy underwater and meet cost constraints
- Lead Electrical/Sensor integration engineer for circuit design, simulation, and testing

Shark Hook Timer rebuild

March 2024 – August 2024

- Individual project focused on prototyping a rebuild of a faulty, discontinued product
 - Rapid prototyping with CAD, custom PCB design, and field pressure testing
 - Worked with customer to specify key field requirements, ensuring the redesigned unit met custom depth, battery life, and durability standards
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RESEARCH & EXPERIENCE

Morphable Biorobotics Laboratory

Laboratory Assistant

September 2025 – Present | Boston, MA

- Managed the mechanical assembly and post processing Soft Balloon Actuators (SBAs) ensuring high-fidelity assembly
- Designed and utilized custom test fixtures to evaluate the structural integrity and range of motion of robotic components
- Collaborated with a cross-functional team to identify and resolve mechanical interference issues during for prototyping

Mote Marine Lab and Aquarium

Underwater Robotics Intern

Mar – Aug 2024 | Sarasota, FL

- Executed field test campaigns for three SLOCUM autonomous gliders and ROVs; performed real-time data analysis to monitor system health and validate performance against mission requirements.
- Led troubleshooting and Root Cause Analysis (RCA) for hardware and communication failures in remote environments, documenting issues to drive system improvements.
- Developed and tested hardware prototypes using Arduino microcontrollers and custom PCB layouts
- Utilized 3D modeling (CAD) to iterate on component designs, focusing on improving durability and ease of maintenance for field deployments

Biosphere 2

Ocean Research Intern

June 2021 – February 2024 | Tucson, AZ

- Applied statistical techniques to interpret large data sets from life support systems and sensors; provided ongoing reports and summaries to the research team to support system performance improvements.
- Collected and analyzed weekly water nutrient samples from water systems (Ocean, Mangroves, & Coral Raceways)
- Performed coral husbandry tasks (cleaning, siphoning, water changes) and assisted in coral fragmentation

University of Arizona, Tropical Reefs Laboratory

Undergraduate Researcher

September 2021 – May 2023 | Tucson, AZ

- Conducted independent paleoclimate research project using coral cores collected from Biosphere 2 ocean
- Analyzed and visualized geochemical tracer trends using MATLAB and Excel; prepared figures and written documentation for lab reports

LEADERSHIP & SERVICE

Engineering Student Ambassador

Graduate Mentor

August 2025 – Present

- Advised incoming Boston University students as they transition to engineering
- Engaged with prospective students and donors on the highlights of the graduate program

Camp Wildcat

Treasurer, Donations Chair

September 2019 – May 2023

- Managed 501(c)(3) club bank account by allocating spending and donations to ensure collections of assets
- Promoted higher education to 4th-8th graders from Title I schools through free weekend-long camping trips

Honors Teaching Assistant

Ocean Sciences and Introduction to Oceanography

Spring 2019 and Spring 2023

- Led weekly study and homework sessions to groups of 25 undergraduate students
- Accompanied class on field work to Southern California for research cruise to collect real-time data and analysis

AWARDS & GRANTS

MTS Microworkshop Scholarship, 2026

Outstanding Senior in Geosciences Award, 2023

Excellence in Undergraduate Research in Geosciences Award, 2023

Best Undergraduate Student Lecture - The University of Arizona GeoDaze Symposium, 2023

AGU Travel grant recipient – \$1000

University of Arizona Travel grant recipient – \$700

CERTIFICATIONS & PROFESSIONAL MEMBERSHIPS

Rutgers University (RUCOOL) SLOCUM Glider Microcredential – June 2026

Institute of Electrical and Electronics Engineers(IEEE, OIEEE), 2026

Marine Technology Society(MTS), 2025

Graduate Women in Science & Engineering(GWISE), 2024

American Geophysical Union(AGU), 2022

PADI Diver Certification – Open Water, 2022

PRESENTATIONS

- The University of Arizona GeoDaze Symposium, Tucson, AZ, March 2023 "*Impact of induced stress on the skeletal geochemistry of corals grown in the Biosphere 2 ocean*" (lecture).
- American Geophysical Union Fall Meeting, Chicago, IL, December 2022 "*Impact of induced stress on the skeletal*

geochemistry of corals grown in the Biosphere 2 ocean" (poster).

- The University of Arizona GeoDaze Symposium, Tucson, AZ, March 2022 "*Impact of induced stress on the skeletal geochemistry of corals grown in the Biosphere 2 ocean*" (poster).